
Is the Reconstruction–Forecasting Tradeoff a Frontier or an Operating Point?

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Abstract

1 Autoencoder based temporal models are usually trained on a weighted sum of a
2 reconstruction loss and a forecasting loss, with the weight tuned per dataset. This
3 assumes the balance between the two objectives is a real frontier. We ask whether it
4 is instead a bad operating point that comes from forcing one latent to do both jobs.
5 On a controlled chaotic benchmark (Lorenz-63 lifted to 30 noisy channels, known
6 intrinsic dimension 3) we split the representation into an unbounded 3-dimensional
7 carrier read only by the forecaster and a sigmoid bounded 8×8 carrier read only
8 by the decoder, trained in three phases so the forecast gradient can never reach the
9 decoder. At matched parameters, matched epochs and matched seeds it reaches a
10 valid prediction time of 1.803 Lyapunov times against 0.619 for the best setting
11 of the knob, 2.91 times longer. No baseline in the study beats climatology at 5
12 Lyapunov times and ours does, with no rollout ever leaving the training range.
13 An ablation separates the two ingredients. We also explain a result that looks
14 impossible: 7 of the baselines reconstruct better than the measured noise floor,
15 which turns out to be copying rather than denoising, and the models that copy most
16 forecast worst.

17 1 Introduction

18 A large body of work trains one encoder to serve two objectives at once. It has to reconstruct the
19 present and forecast the future. The standard way to do this is a weighted loss, $\mathcal{L} = (1 - \phi)\mathcal{L}_{\text{rec}} +$
20 $\phi\mathcal{L}_{\text{fct}}$, with ϕ tuned per dataset. Coupled Attention Networks [Xia et al., 2024] call their weights
21 “a tradeoff between the prediction and reconstruction models”. CAAE [Xie et al., 2026] names
22 the “objective mismatch between training for reconstruction fidelity and testing for discriminative
23 decisions”. Reconstruction plus prediction hybrids fuse both objectives onto one shared latent, some
24 with explicit weights [Xiao et al., 2021] and some by plainly summing the two losses [Cai et al., 2023].
25 SSP [Lu et al., 2026] chases the same separation by spectral truncation, “decoupling reconstruction
26 fidelity from rollout regularity”. In all of this work the tradeoff is something you tune. Almost nobody
27 asks whether the tuning is needed at all.

28 The ingredients for asking already exist, but separately. Bounded latents are routine in representation
29 learning [Mentzer et al., 2024]. Koopman style forecasters constrain the latent *dynamics* to buy long
30 horizon stability [Azencot et al., 2020, Liu et al., 2023]. Neither line splits the two objectives across
31 separate carriers, and the Koopman forecasters still run a single latent through both.

32 **What we are not claiming.** Rate distortion theory [Shannon, 1959, Berger, 1971] proves a frontier
33 between rate and distortion for a source, and the information bottleneck [Tishby et al., 2000] extends

34 this to predictive representations. We are not claiming to beat any such bound. Rate distortion
 35 constrains distortion *at a fixed rate* and says nothing about which point inside the achievable region
 36 an architecture lands on. Our claim is only the second one. At fixed capacity and fixed compute, a
 37 shared latent under a weighted loss does not reach the best available point, and separating the carriers
 38 moves it there.

39 **Contributions.** (i) A capacity matched and epoch matched test of whether the tradeoff is a frontier
 40 or an operating point, on a system of known intrinsic dimension. (ii) A three phase schedule with a
 41 bounded and an unbounded carrier that reaches 2.91 times the forecast horizon of the best weighted
 42 loss setting, while reconstructing at 1.012 times the measured noise floor and never leaving the
 43 training range. (iii) An explanation of why wide latent baselines appear to reconstruct below the
 44 noise floor.

45 2 Method

46 Write $x_t \in \mathbb{R}^n$ for the observation at time t . The model has six parts. An encoder E , a bounded
 47 teacher r , a decoder D , a compressor f , a bounded map m , and a forecaster g . The inference path is

$$x_t \xrightarrow{E} h_t \xrightarrow{f} b_t \in \mathbb{R}^k \xrightarrow{m} C_t \in [0, 1]^{8 \times 8} \xrightarrow{D} \hat{x}_t, \quad b_{t+1} = g(b_t). \quad (1)$$

48 The carrier b is unbounded and only the forecaster reads it. The carrier C is bounded and only the
 49 decoder reads it. C stays inside $[0, 1]$ because m ends in a sigmoid, and that holds for a rollout of any
 50 length however far b drifts, because it is a property of the function class and not of the optimiser.

51 **Three phase training.** The two carriers are separated by the schedule, not only by the wiring. Each
 52 phase freezes everything it does not train.

53 *Phase 1* trains E , r and D as a bounded autoencoder, using $E(x_t) = h_t$, $r(h_t) = C_t$ and $D(C_t) = \hat{x}_t$
 54 with $\mathcal{L}_1 = \text{MSE}(\hat{x}_t, x_t)$. The decoder never sees the narrow carrier during its own training.

55 *Phase 2* freezes E , r and D , and trains f , g and m with $f(h_t) = b_t$, $g(b_t) = b_{t+1}$ and $m(b_{t+1}) =$
 56 $\hat{C}_{t+1}$. The loss lives in C space against the frozen phase 1 code, $\mathcal{L}_2 = \text{MSE}(C_{t+1}, \hat{C}_{t+1})$. The
 57 decoder is not used at all here. The target $C_{t+1} = r(h_{t+1})$ is a fixed tensor, so f cannot escape by
 58 collapsing. A loss in b space against the encoder’s own b would collapse instead, because a constant
 59 f and an identity g drive it to zero while learning nothing.

60 *Phase 3* freezes E , f , g and r , and fine tunes m and D with $m(b_t) = \hat{C}_t$ and $D(\hat{C}_t) = \hat{x}_t$, under
 61 $\mathcal{L}_3 = \text{MSE}(\hat{C}_t, C_t) + \text{MSE}(\hat{x}_t, x_t)$.

62 The decoupling is exact and not approximate. In phase 3 the forecaster is frozen, so $\partial \mathcal{L}_3 / \partial \theta_g \equiv 0$. In
 63 phase 2 the decoder is frozen and never even runs. No gradient from a forecast objective ever reaches
 64 D .

65 3 Experimental setup

66 **Data.** Lorenz-63 [Lorenz, 1963] at $dt = 0.01$, lifted through a frozen random tanh map to 30
 67 channels with additive noise, so the intrinsic dimension is known to be 3. We use 8 training trajectories
 68 of 20,000 steps each, split 70/15/15 in time and pooled, giving 112,000 timesteps. Every scaling
 69 constant is fit on that pooled training portion only. A further 32 trajectories are held out completely
 70 and never split, and all reported numbers come from windows cut from those. The measured noise
 71 floor is 0.002492 in the final standardised scale.

72 **Metrics.** Reconstruction MSE on held out windows, and its ratio to the noise floor. Valid prediction
 73 time (VPT), the lead time at which normalised error first crosses 0.4, in Lyapunov times [Vlachas
 74 et al., 2020]. Normalised error over the full 5 Lyapunov time horizon. Divergence, the fraction of
 75 rollouts that leave the training range. One Lyapunov time is 110.4 steps.

Table 1: Held out results, sorted by forecast horizon. Lower is better for reconstruction MSE, error at 5 LT and divergence. Higher is better for VPT. Every trained row has 124,482 parameters and 60 epochs per parameter group. The $\times$ floor column is reconstruction MSE divided by the measured noise floor. MAE and ACC horizon come from the long horizon pass described in Section 4. ACC horizon is the lead time at which anomaly correlation drops below 0.6.

model	seeds	recon MSE	$\times$ floor	VPT (LT)	err @ 5LT	diverg.	MAE @ 1LT	ACC hor.
Ours (three-phase)	3	0.00252	1.012	1.803 ± 0.073	0.822	0.0%	0.151	2.99
Ours (one-stage)	1	0.00253	1.016	1.114	0.971	0.0%	0.212	2.45
$\phi=0.1$	3	0.00160	0.643	0.619 ± 0.009	1.195	1.0%	0.590	1.18
$\phi=0.5$	3	0.00217	0.871	0.444 ± 0.051	5.957	99.0%	1.120	0.72
$\phi=0.25$	3	0.00188	0.753	0.441 ± 0.028	2.052	27.6%	1.015	0.72
AEGRU+sigmoid (D)	3	0.00227	0.910	0.420 ± 0.015	1.236	2.6%	1.054	0.72
PINN (true physics)	3	0.00419	1.682	0.405 ± 0.004	1.145	18.8%	0.516	1.18
$\phi=1$	1	0.00372	1.492	0.371	4.995	100.0%	1.185	0.54
$\phi=0$	1	0.00135	0.543	0.018	94.916	100.0%	21.920	0.09
persistence	—	—	—	0.036	1.369	0.0%	1.354	0.18
climatology	—	—	—	0.000	1.002	0.0%	0.862	0.00

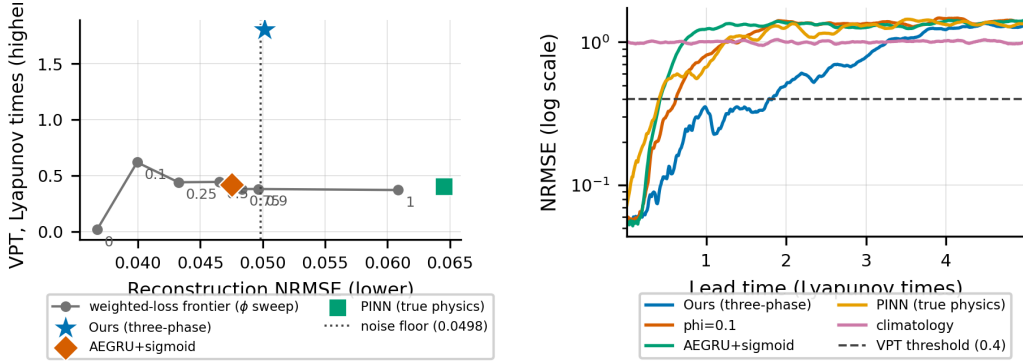


Figure 1: Left: reconstruction against forecast horizon. The ϕ sweep traces the weighted loss frontier and our split sits well above all of it. Right: normalised error against lead time on a log scale. Ours is the only curve that stays under climatology across the whole horizon.

Protocol. Every model gets 124,482 trainable parameters, matched by bisecting the baseline hidden width, which lands within 0.36%. Every parameter group gets 60 passes over its own training data. Epochs are used instead of wall clock because epoch counts reproduce on any machine and wall clock does not. Three seeds were stated. Baselines are the weighted loss autoencoder plus GRU swept over ϕ , a sigmoid bounded shared latent control that isolates boundedness from the split, and a physics informed model handed the exact Lorenz equations.

4 Results

The forecast gap is large. Our split reaches VPT 1.803 Lyapunov times, 199 steps, against 0.619 for the best knob setting at $\phi=0.1$, so 2.91 times longer. Turning the knob does not close the gap, because VPT peaks at $\phi=0.1$ and then falls. At 5 Lyapunov times ours reports 0.822 against 1.002 for climatology, and no baseline in the study beats predicting the mean at all.

What the architecture buys and what the schedule buys. The same six modules trained end to end in one stage reach VPT 1.114 against 1.803, so the schedule is worth 1.62 times on horizon. It buys nothing on reconstruction (1.016 against 1.012 times the floor) or on stability (both 0%), which the architecture already supplies. Both ingredients matter, because the one stage run alone beats the best knob setting by 1.80 times.

92 **Forecast skill agrees with VPT, on a different axis.** VPT reads one error threshold, so we also
 93 profiled MAE, anomaly correlation (ACC), skill against persistence and forecast variance out to 50
 94 Lyapunov times (Appendix D). At one Lyapunov time ours reports MAE 0.151 against 0.516 for the
 95 best baseline, and ACC stays above 0.6 out to 2.99 Lyapunov times against 1.18, a factor of 2.54.
 96 ACC scores whether the pattern is right and VPT whether the error is small, so this is an independent
 97 check. Ours also holds variance ratio 1.036 at 50 Lyapunov times while 6 knob settings blow past
 98 twice the true variance.

99 **Stability is a guarantee, not a tuning result.** Over 5,500 free running steps no entry of C ever
 100 leaves $[0, 1]$ while $|b|$ reaches 11.5, which is the bounded carrier doing what the construction promises.
 101 Attractor fidelity agrees, with Wasserstein 0.0302 against 0.0566 for the PINN and 1.095 for $\phi=0.1$.

102 5 Why some baselines land under the noise floor

103 7 of the trained baselines reconstruct *below* the measured noise floor. That looks impossible, but it
 104 is not. The floor is the MSE you get by outputting the clean signal, while reconstruction is scored
 105 as $\text{MSE}(\hat{x}, x)$ where x is the *noisy* observation and the model takes x as its input. So the floor was
 106 never a lower bound here. The identity map settles it. Set $\hat{x} = x$ and the score is exactly zero.

107 The clean signal lies on a k dimensional manifold because it is a fixed function of the Lorenz state,
 108 while the noise spreads over all n channels. A model with a d dimensional bottleneck can carry d of
 109 those noise directions through instead of discarding them, so the achievable minimum is

$$\text{MSE}_{\min} = \frac{n-d}{n} \times \text{floor}, \quad (2)$$

110 not the floor itself. Figure 2 in the appendix checks this against optimal linear rank d reconstruction
 111 of the same held out windows. At the baseline latent width $d=16$ the measured value is 0.466 against
 112 a predicted 0.467.

113 We measure the copying directly. Perturb the input a little and see how much survives to the output.
 114 A pure identity gives 1, a rank d projection gives d/n , and a model passing only the signal gives k/n .
 115 At $\phi=0$ we measure 0.517 against 0.533 for a rank 16 copy, while ours measures 0.113 against 0.100
 116 for a pure denoiser.

117 Two things follow. First, the reconstruction column is not comparable across bottleneck widths,
 118 because wide models have a lower achievable floor. Normalised by each model’s own bound, ours is
 119 best of the trained models at 1.125 against 1.165 (Table 2). Second, copying and forecasting conflict.
 120 Passthrough falls from 0.517 at $\phi=0$ to 0.116 at $\phi=0.5$ and 0.097 at $\phi=1$, and $\phi=0$, which copies
 121 most and looks best on reconstruction, is the worst forecaster in the study at VPT 0.018 with 100%
 122 divergence. The reconstruction win and the forecast collapse are one fact measured twice.

123 6 Limitations

124 **One system.** Lorenz-63, one lift, one noise level. **Uneven budget.** m and D train in two phases
 125 and see 120 passes against 60, and the one stage ablation gets 60 epochs against three phases of 60,
 126 so part of the schedule gap may be compute. **Uneven seeds.** Single seed rows report zero spread.
 127 **Constructed baselines.** The ϕ sweep and the sigmoid control are the balancing recipe in its plainest
 128 form, not named methods. **Fixed capacity.** The bounded carrier is a fixed 8×8 grid we never varied.

129 7 Conclusion

130 At matched parameters, matched epochs and matched seeds, separating a bounded decoder facing
 131 carrier from an unbounded forecaster facing one reaches an operating point no setting of the weighted
 132 loss knob reaches: 2.91 times its forecast horizon, reconstruction at 1.012 times the noise floor, and
 133 no rollout leaving the training range. Read together with the copying result, the tradeoff looks less
 134 like a frontier and more like a symptom of asking one carrier to do two jobs.

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167 A Capacity normalised reconstruction

168 Reconstruction MSE is scored against the noisy observation, so a model with a d dimensional
 169 bottleneck has achievable minimum $(n - d)/n$ times the floor rather than the floor itself. Table 2
 170 divides each model’s score by its own bound. On this view ours is the best of the trained models. The
 171 capacity matched diagnostic, a $\phi=0$ baseline with its latent cut from 16 to 3, is included to show that
 172 the effect tracks width and not architecture. It lands at 1.058 times its own optimum, which is close
 173 to ours, and its forecast collapses in exactly the way $\phi=0$ does.

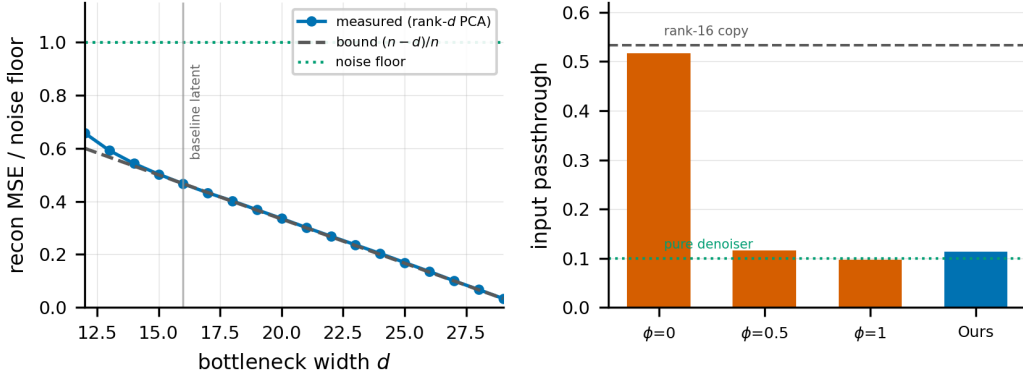


Figure 2: Left: optimal linear rank d reconstruction of the held out windows tracks the capacity bound $(n - d)/n$ almost exactly, and sits far under the noise floor for any wide bottleneck. Right: measured passthrough, the fraction of a random input perturbation that survives to the output. $\phi=0$ behaves like a rank 16 copy of its input. Ours sits at the pure denoiser value, because a 3 dimensional carrier has no spare width to copy with.

Table 2: Reconstruction against each model’s own capacity bound. d is the bottleneck width. The bound column is $(n - d)/n$.

model	d	$\times$ floor	bound	$\times$ own optimum
$\phi=0$, latent=3	3	0.952	0.900	1.058
Ours	3	1.012	0.900	1.125
$\phi=0$	16	0.543	0.467	1.165
$\phi=0.1$	16	0.643	0.467	1.377
$\phi=0.25$	16	0.753	0.467	1.614
$\phi=0.5$	16	0.871	0.467	1.867
PINN (true physics)	3	1.682	0.900	1.869
AEGRU+sigmoid	16	0.910	0.467	1.950
$\phi=0.75$	16	0.941	0.467	2.016
$\phi=0.9$	16	0.992	0.467	2.126
Ours (one-stage)	16	1.016	0.467	2.178
$\phi=1$	16	1.492	0.467	3.196
PINN (rho=26)	3	5.054	0.900	5.615

174 B Input passthrough

175 Passthrough is measured by perturbing the input with small isotropic noise and recording the fraction
 176 of that perturbation which survives to the output, averaged over 8 draws. A pure identity gives 1. A
 177 rank d projection gives d/n . A model whose output depends only on the 3 dimensional signal gives
 178 k/n .

Table 3: Measured passthrough against the two reference behaviours.

model	d	measured	rank d copy	pure denoiser
Ours	3	0.113	0.100	0.100
$\phi=0$	16	0.517	0.533	0.100
$\phi=0.5$	16	0.116	0.533	0.100
$\phi=1$	16	0.097	0.533	0.100

179 C Long horizon attractor statistics

180 Free running for 5,500 steps, which is 50 Lyapunov times, on 64 held out windows. Wasserstein
181 distance is averaged over the observed channels and the max column is the worst channel.

Table 4: Attractor fidelity at 50 Lyapunov times.

model	Wasserstein	worst channel	log spectrum error
Ours	0.0302	0.0513	2.135
$\phi=0$	671.3238	1804.0989	10.853
$\phi=0.1$	1.0948	1.5373	2.615
$\phi=0.25$	51.8543	93.0053	7.106
$\phi=0.5$	68.8035	110.5312	6.890
$\phi=0.75$	69.4604	106.9682	6.872
$\phi=0.9$	68.1392	103.4794	6.821
$\phi=1$	63.8754	80.8080	6.696
AEGRU+sigmoid	0.5898	0.9128	3.142
PINN (true physics)	0.0566	0.1059	1.977
PINN ($\rho=26$)	0.1058	0.2487	1.808
Ours (one-stage)	0.0770	0.1169	2.038
$\phi=0$, latent=3	121.9196	353.8474	6.646
persistence	0.1823	0.2906	31.344
climatology	0.8458	0.9088	31.357

182 D Full lead time profile

183 Free running for 5,500 steps on 64 held out windows, scored at seven lead times. MAE is in
184 standardised observation units. ACC is anomaly correlation against the training climatology, where
185 1 is perfect and 0 is no better than predicting the mean. Skill is measured against persistence, so
186 0 means no better than assuming nothing changes and negative means worse. The variance ratio
187 is forecast standard deviation over true standard deviation, so 1 is the right amplitude, below 1 is
188 collapsing toward the mean and above 1 is blowing up.

189 The variance ratio is worth reading alongside divergence, because the two disagree in an informative
190 way. $\phi=0.1$ holds amplitude 1.116 at 50 Lyapunov times and never leaves the training range, yet
191 it scores worst of all on attractor fidelity. It neither collapses nor explodes. It simply decorrelates,
192 which is a failure only ACC and the attractor statistics can see.

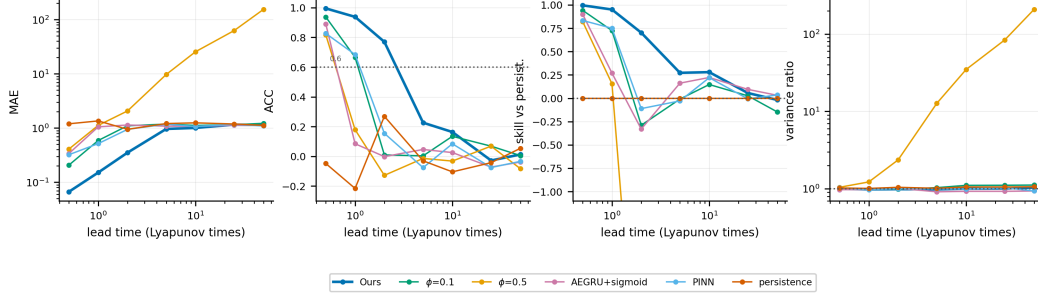


Figure 3: Lead time profile for a representative subset. Ours is the only model whose anomaly correlation is still above 0.6 past one Lyapunov time, and it holds the right variance for the whole run while $\phi=0.5$ grows past $100\times$ the true amplitude.

Table 5: MAE at each lead time, in Lyapunov times.

model	0.5	1	2	5	10	25	50
Ours	0.067	0.151	0.352	0.963	1.007	1.159	1.135
$\phi=0$	9.583	21.920	48.390	130.782	267.229	676.615	1345.309
$\phi=0.1$	0.207	0.590	1.102	1.183	1.118	1.152	1.223
$\phi=0.25$	0.387	1.015	1.318	2.530	6.898	42.170	145.840
$\phi=0.5$	0.410	1.120	2.069	9.734	25.594	63.008	154.900
$\phi=0.75$	0.477	1.102	2.217	7.634	20.706	68.241	156.402
$\phi=0.9$	0.612	1.010	1.830	6.855	20.179	67.228	149.962
$\phi=1$	0.663	1.185	1.946	7.039	20.577	63.528	138.395
AEGRU+sigmoid	0.339	1.054	1.138	1.084	1.078	1.126	1.118
PINN	0.323	0.516	0.946	1.180	1.067	1.174	1.094
PINN ($\rho=26$)	0.663	0.770	1.062	1.068	1.137	1.142	1.239
$\phi=0$, lat. 3	1.210	2.268	7.751	26.024	53.691	124.045	238.806
persistence	1.199	1.354	0.949	1.211	1.248	1.188	1.121
climatology	0.858	0.862	0.838	0.851	0.831	0.837	0.830

Table 6: Anomaly correlation at each lead time.

model	0.5	1	2	5	10	25	50
Ours	0.995	0.938	0.770	0.226	0.163	-0.028	0.012
$\phi=0$	0.025	-0.011	-0.011	-0.009	0.007	0.009	-0.075
$\phi=0.1$	0.938	0.666	0.009	0.003	0.135	0.068	0.005
$\phi=0.25$	0.818	0.234	-0.108	-0.018	-0.074	-0.027	-0.059
$\phi=0.5$	0.818	0.179	-0.128	-0.014	-0.032	0.068	-0.083
$\phi=0.75$	0.774	0.121	0.047	0.065	0.085	0.044	-0.025
$\phi=0.9$	0.677	0.161	0.031	-0.007	-0.116	0.036	0.037
$\phi=1$	0.608	-0.024	-0.000	0.031	-0.018	0.051	0.070
AEGRU+sigmoid	0.890	0.085	-0.002	0.046	0.025	-0.073	-0.038
PINN	0.826	0.685	0.155	-0.076	0.084	-0.075	-0.034
PINN ($\rho=26$)	0.516	0.426	0.075	0.070	-0.030	0.039	-0.181
$\phi=0$, lat. 3	0.000	0.006	-0.070	0.012	0.096	0.009	0.048
persistence	-0.048	-0.216	0.269	-0.030	-0.105	-0.043	0.054
climatology	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Table 7: Skill against persistence at each lead time. Large negative values mean the forecast has left the attractor entirely.

model	0.5	1	2	5	10	25	50
Ours	1.00	0.95	0.70	0.27	0.28	0.06	-0.02
$\phi=0$	-65.08	-290.19	-2438.85	-12277.46	-50560.65	-346024.43	-1520453.07
$\phi=0.1$	0.94	0.73	-0.29	-0.01	0.15	0.03	-0.15
$\phi=0.25$	0.83	0.34	-0.71	-10.60	-168.53	-3437.71	-32478.57
$\phi=0.5$	0.82	0.16	-3.74	-77.74	-523.31	-3234.92	-21936.63
$\phi=0.75$	0.79	0.20	-5.41	-59.98	-368.09	-3887.38	-23140.92
$\phi=0.9$	0.72	0.39	-2.61	-39.15	-344.46	-3921.71	-21488.58
$\phi=1$	0.67	0.22	-3.68	-44.36	-347.72	-3289.58	-18210.52
AEGRU+sigmoid	0.90	0.27	-0.33	0.16	0.22	0.09	0.03
PINN	0.83	0.75	-0.11	-0.03	0.22	-0.01	0.04
PINN ($\rho=26$)	0.55	0.54	-0.18	0.16	0.10	0.07	-0.16
$\phi=0$, lat. 3	-0.32	-2.97	-58.98	-417.12	-1718.93	-11378.65	-52418.89
persistence	0.00	0.00	0.00	0.00	0.00	0.00	0.00
climatology	0.53	0.59	0.35	0.52	0.57	0.55	0.50

Table 8: Variance ratio at each lead time. 1 is the right amplitude.

model	0.5	1	2	5	10	25	50
Ours	1.003	0.981	0.989	0.982	1.003	1.010	1.036
$\phi=0$	11.706	26.214	59.911	157.045	336.946	857.199	1714.307
$\phi=0.1$	0.999	1.010	0.987	1.034	1.106	1.109	1.116
$\phi=0.25$	1.015	1.052	1.160	4.798	19.766	86.891	254.878
$\phi=0.5$	1.044	1.225	2.360	12.728	34.817	84.136	208.598
$\phi=0.75$	1.006	1.099	3.003	11.289	29.311	92.334	215.395
$\phi=0.9$	0.931	0.881	2.156	9.097	28.235	92.871	207.568
$\phi=1$	0.893	0.939	2.475	9.698	28.398	85.149	191.416
AEGRU+sigmoid	0.972	0.979	1.014	0.914	0.927	0.926	0.936
PINN	1.027	0.974	1.004	0.998	0.992	1.030	0.936
PINN ($\rho=26$)	0.992	0.980	0.975	0.945	1.019	1.061	0.987
$\phi=0$, lat. 3	1.331	2.937	9.449	29.534	63.396	158.084	323.448
persistence	1.023	1.010	1.044	1.014	1.053	1.054	1.061
climatology	0.000	0.000	0.000	0.000	0.000	0.000	0.000

193 E Reproducibility

194 All numbers in this paper are produced by code from saved checkpoints, including every ratio quoted
195 in the text. The three phase schedule runs 60 epochs per phase on 3 seeds. Phase 1 takes 58 seconds,
196 phase 2 takes 945 seconds and phase 3 takes 523 seconds on a single desktop CPU with 8 threads.
197 Metrics are snapshotted at epochs 5, 10, 20, 40, 60 so the whole compute scaling curve comes from
198 one run rather than from separate runs at different budgets.